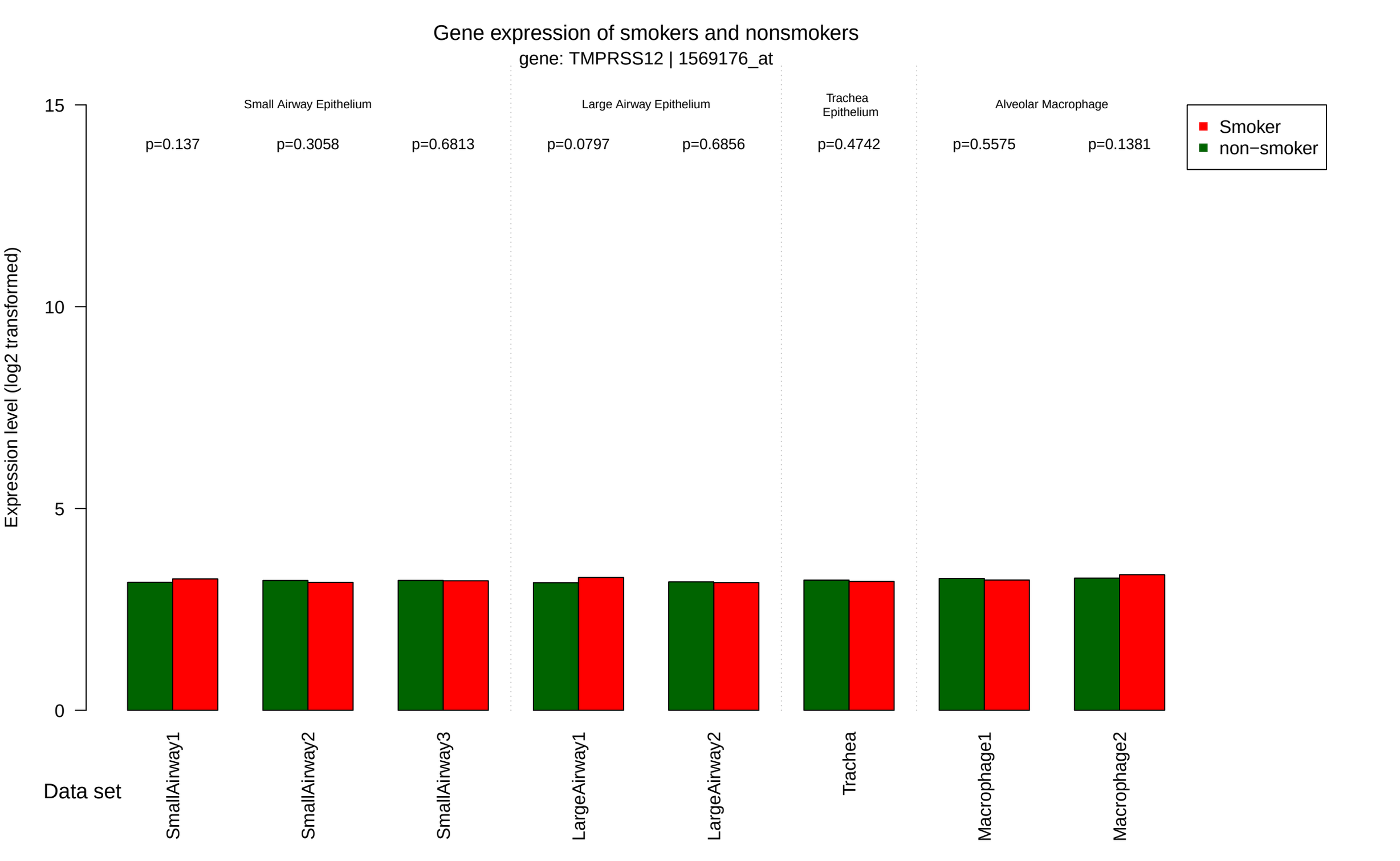
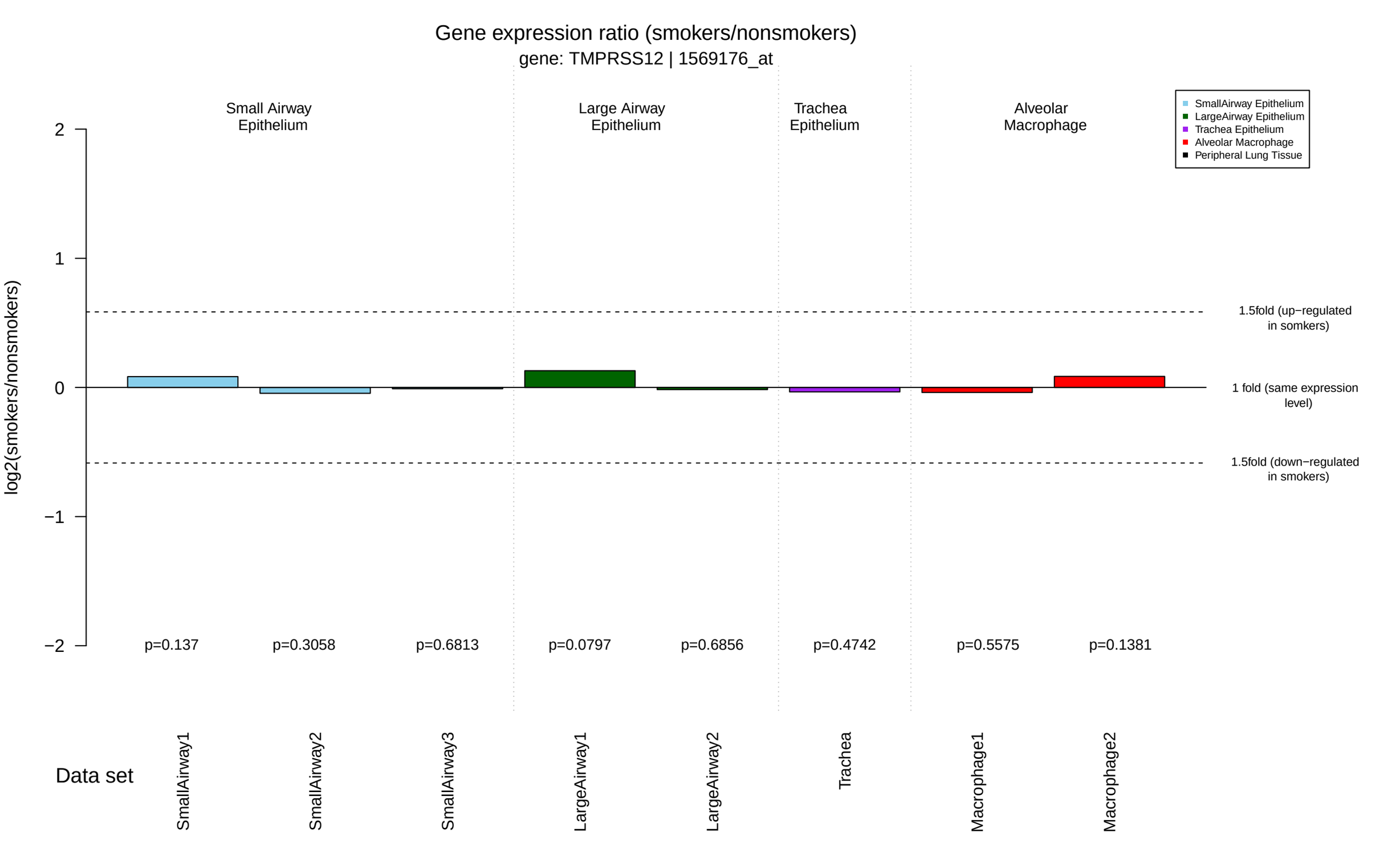
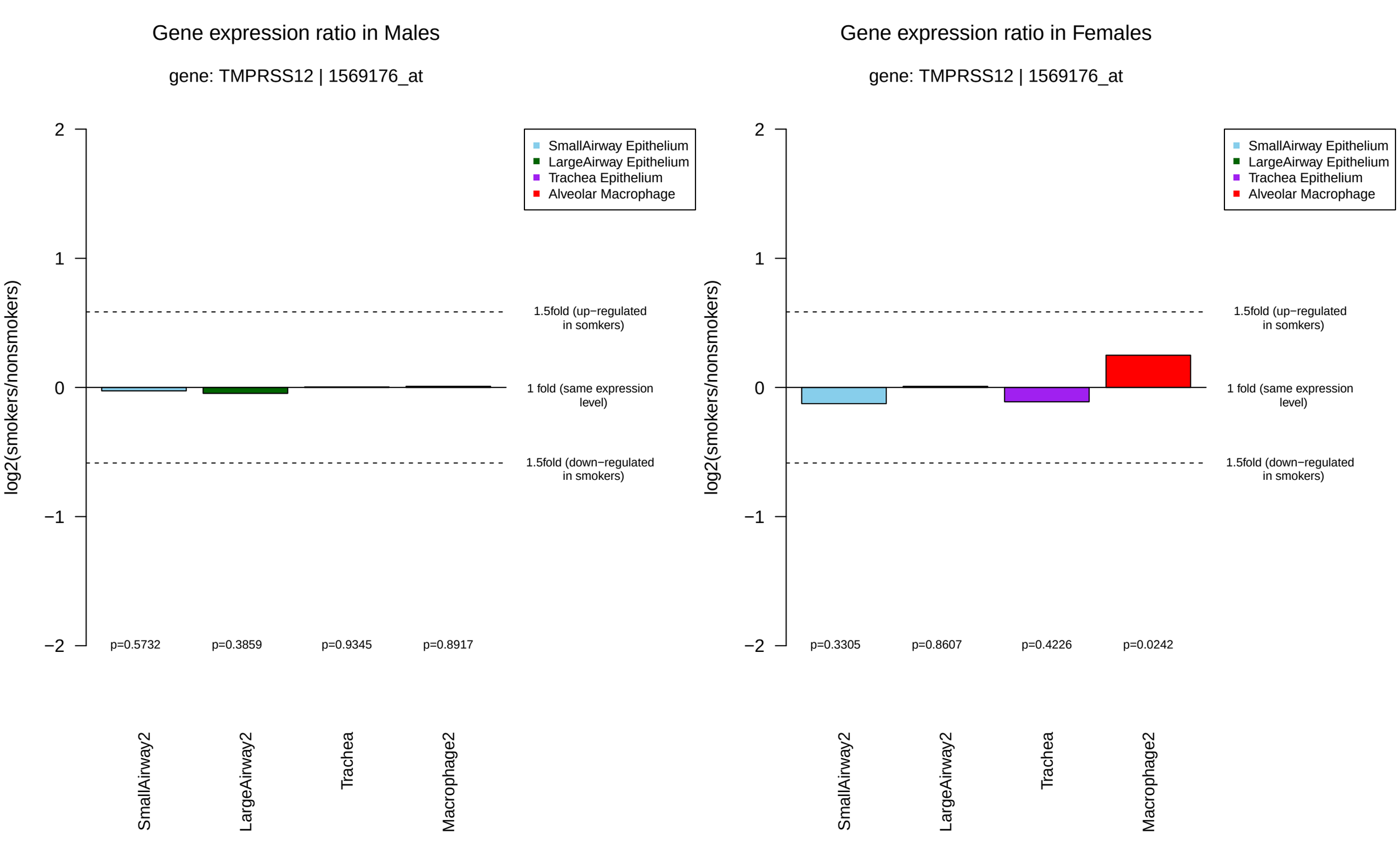
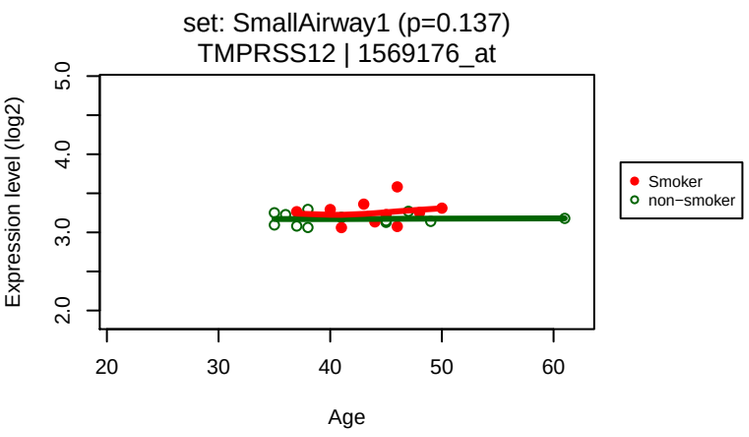
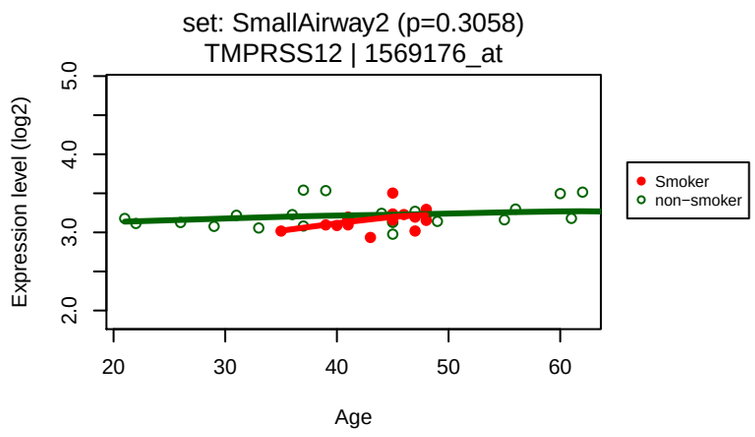
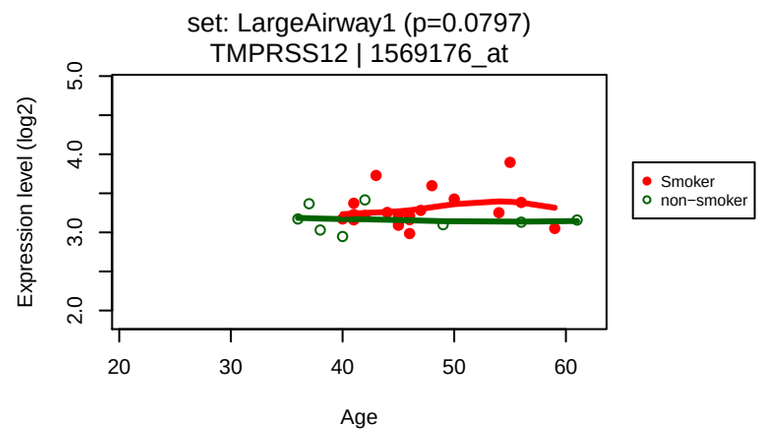

Gene expression of smokers and nonsmokers
gene: TMPRSS12 | 1569176_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.137
p=0.3058
p=0.6813
p=0.0797
p=0.6856
p=0.4742
p=0.5575
p=0.1381
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

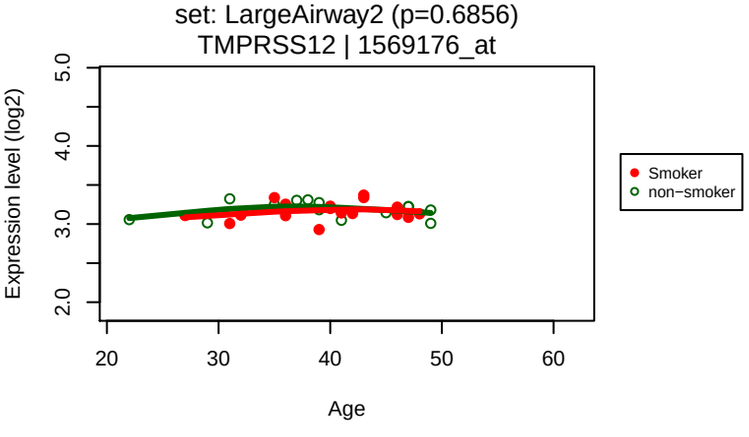

Gene expression ratio (smokers/nonsmokers)
gene: TMPRSS12 | 1569176_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.137
p=0.3058
p=0.6813
p=0.0797
p=0.6856
p=0.4742
p=0.5575
p=0.1381
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

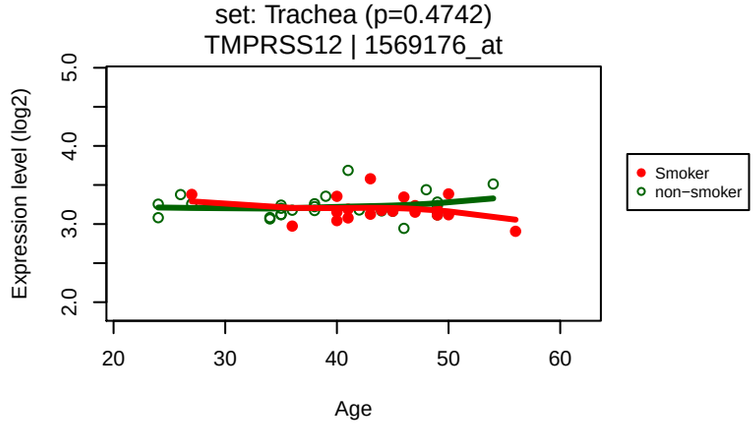

Gene expression ratio in Males
gene: TMPRSS12 | 1569176_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5732
p=0.3859
p=0.9345
p=0.8917
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMPRSS12 | 1569176_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3305
p=0.8607
p=0.4226
p=0.0242
SmallAirway2
LargeAirway2
Trachea
Macrophage2

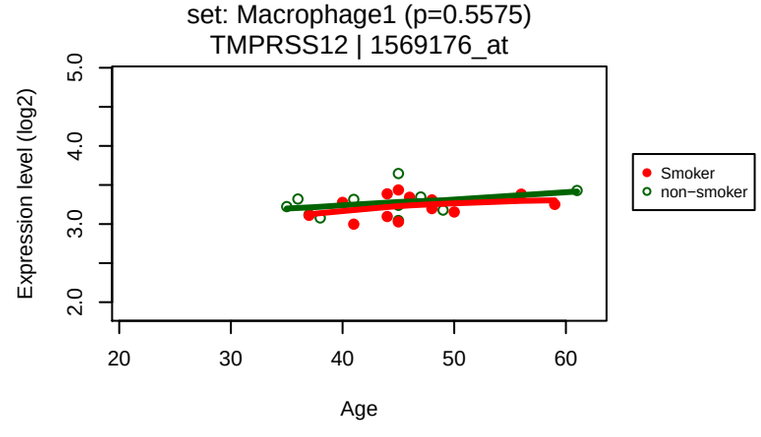

set: SmallAirway1 (p=0.137)
TMPRSS12 | 1569176_at
Expression level (log2)
Age
Smoker
non−smoker

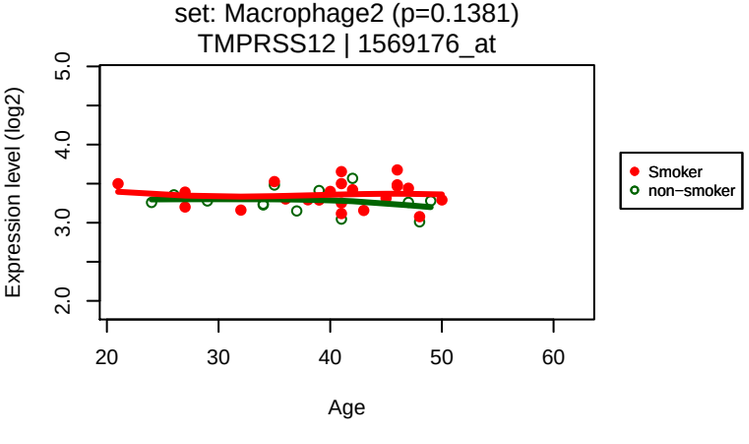

set: SmallAirway2 (p=0.3058)
TMPRSS12 | 1569176_at
Expression level (log2)
Age
Smoker
non−smoker

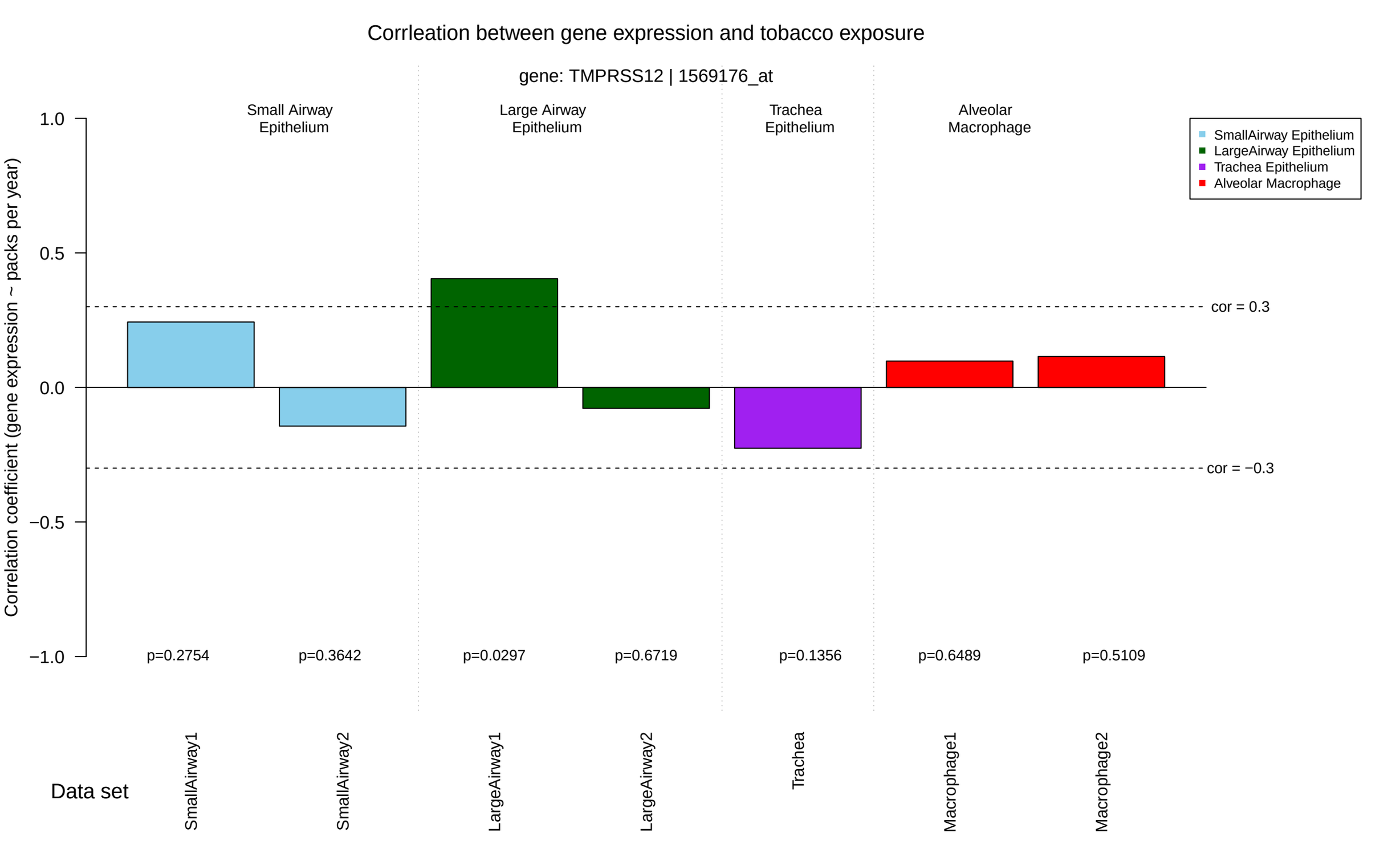

set: LargeAirway1 (p=0.0797)
TMPRSS12 | 1569176_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6856)
TMPRSS12 | 1569176_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4742)
TMPRSS12 | 1569176_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5575)
TMPRSS12 | 1569176_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1381)
TMPRSS12 | 1569176_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMPRSS12 | 1569176_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2754
p=0.3642
p=0.0297
p=0.6719
p=0.1356
p=0.6489
p=0.5109
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2